

LLM for Teens

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■ Session 1 Recap

What We Covered in Session 1:

What's an LLM? We learned what a Large Language Model is and how it's different from traditional NLP.

LLM Structure: We explored how LLMs use transformers, attention, and tokens.

Key Concepts: We talked about terms like temperature, top-k, and top-p to understand how LLMs generate responses.

Intro to RAG: We introduced Retrieval-Augmented Generation (RAG) and how it helps LLMs access extra knowledge.

■ Session 1 Recap

session 1 quick review

- llm: next-word predictor
- why llms matter: smart because trained on millions of examples
- llm vs nlp: nlp uses rules, llm learns from examples
- tokens: text pieces
- embeddings: meaning as numbers
- attention: focus on important words
- transformer: layers + attention
- temperature: creativity control
- top-k: choose from top k words
- top-p: choose from the most likely words
- rag: add knowledge without training

■ What embeddings are

Embeddings basics = turning text into numbers that represent meaning.

They place words or sentences in a vector space where similar meanings are close and different meanings are far.

■ Why embeddings matter

Embeddings basics = turning text into numbers that represent meaning.

They place words or sentences in a vector space where similar meanings are close and different meanings are far.

Meaning = the idea behind the text, not the exact words.

■ How embeddings are created

Embeddings are created by passing text through an LLM's encoder to get a vector that represents its meaning.

They come from the internal hidden layers of the model after it processes the text.

■ Cosine similarity

Cosine similarity = how close two embedding vectors point in the same direction.

High score → meanings are similar

Low score → meanings are different

■ Building a document embedding store

Building a document embedding store = creating a list of documents and saving their embedding vectors so you can search them later.

Steps:

- collect documents
- generate an embedding for each
- save vectors in a list or database
- use them for similarity search

■ Semantic search algorithm

Semantic search algorithm = find the most relevant document by comparing embeddings.

Steps:

- embed the query
- embed all documents
- compute cosine similarity
- pick the highest score

■ Limitations of embeddings

- loss of detail — small meaning differences get removed
- context changes meaning — meaning shifts when the text around it changes
- domain mismatch — works worse on special fields like medical or legal
- static knowledge — does not learn new facts unless retrained
- similarity mistakes — sometimes shows wrong “similar” results

■ How embeddings connect to RAG

- embeddings find meaning — they turn text into vectors so the system can compare meanings
- query → vector — your question is turned into an embedding
- document → vector — all documents are also turned into embeddings
- semantic search — cosine similarity finds the closest document vectors
- RAG uses those results — the top matching documents are passed to the LLM as context
- LLM answers using the found text — the model uses the text it received to give a correct answer

Assignment

- Create a RAG system using the `python_questions_dataset`. Load the questions and answers, merge them by Id, embed the questions, find the closest match to the user query, and return the matched question and its answer.

Summary

- ✓ What embeddings are
- ✓ Why embeddings matter
- ✓ How embeddings are created
- ✓ Cosine similarity
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- ✓ Semantic search algorithm
- ✓ Coding: embedding + cosine similarity
- ✓ Coding: mini semantic search engine
- ✓ Limitations of embeddings
- ✓ How embeddings connect to RAG

Thank You

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